

MODULE 14

SQL Transactions and ACID Properties

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Transactions and ACID Properties

What is a Transaction?

Sequence of SQL operations treated as one unit.

All succeed (COMMIT) or all fail (ROLLBACK).

Classic example: bank transfer must be atomic.

COMMIT and ROLLBACK

```
START TRANSACTION;

UPDATE accounts SET balance = balance - 1000 WHERE id = 1;
UPDATE accounts SET balance = balance + 1000 WHERE id = 2;

COMMIT; -- save changes
-- OR
ROLLBACK; -- undo everything

SAVEPOINT sp1;
ROLLBACK TO sp1;
```

ACID

Atomicity: all or nothing

Consistency: database moves between valid states

Isolation: transactions do not see each other's work

Durability: committed data survives crashes

Isolation Levels

```
READ UNCOMMITTED: dirty reads possible
READ COMMITTED: only committed data visible
REPEATABLE READ: stable results per query (MySQL default)
SERIALIZABLE: fully isolated, strictest

SET TRANSACTION ISOLATION LEVEL REPEATABLE READ;
```

Transaction Procedure

```
DELIMITER //
CREATE PROCEDURE Transfer(IN fid INT, IN tid INT, IN amt DECIMAL)
BEGIN
  DECLARE CONTINUE HANDLER FOR SQLEXCEPTION ROLLBACK;
  START TRANSACTION;
  UPDATE accounts SET balance = balance - amt WHERE id = fid;
  UPDATE accounts SET balance = balance + amt WHERE id = tid;
```

```
COMMIT;  
END //
```